STANDARD GRASS SEED MIXES FOR

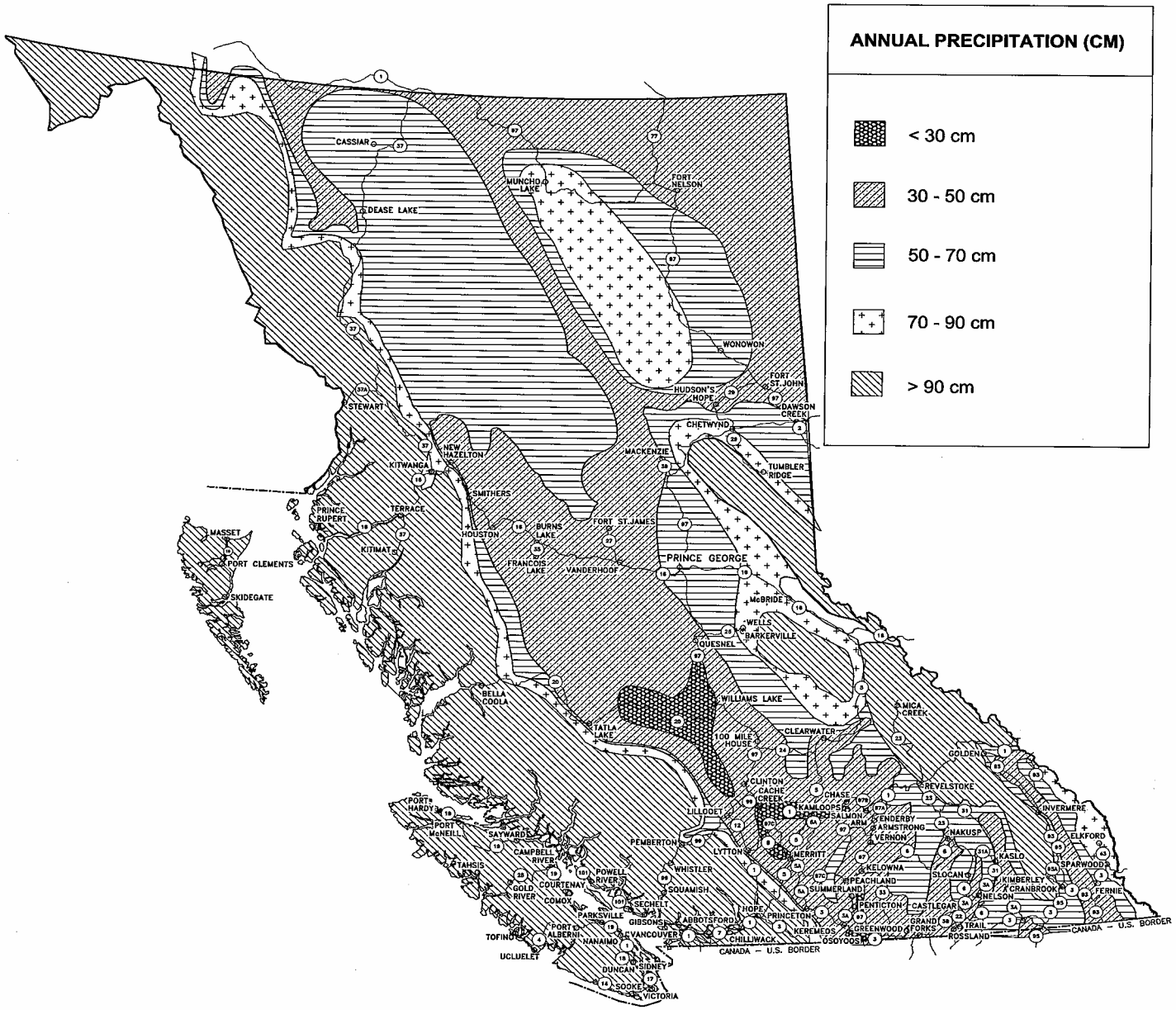
REVEGETATION OF BRITISH COLUMBIA HIGHWAY ROADSIDES (BY WEIGHT)

CLIMATIC AREA	STANDARD MIXES (by weight)	APPLICATION
South Coast	**Vancouver Island / Coast Mix** Perennial Ryegrass 26% Creeping Red Fescue 24% Alsike Clover 14% Hard Fescue 13% White Clover 9% Timothy 8% Canada Bluegrass 4% Redtop 2%	General seeding coastal locations where mean annual precipitation is > 90 cm. Fertilizer: 16-32-6
	Interior Forestland Mix Intermediate Wheatgrass 32% Alfalfa ("Rambler") 20% Perennial Ryegrass 15% Annual Ryegrass 15% Hard Fescue 10% White Dutch Clover 5% Canada Bluegrass 2% Redtop 1%	General seeding inland where mean annual precipitation is >50 cm. Fertilizer: 16-32-6
	Interior Dryland Mix Crested Wheatgrass 40% Tall Wheatgrass 25% Slender Wheatgrass 20% Hard Fescue 15%	General seeding inland where mean annual precipitation is < 30 cm. Fertilizer: 16-32-6
Thompson – Okanagan	**Interior Forestland Mix** Intermediate Wheatgrass 32% Alfalfa ("Rambler") 20% Perennial Ryegrass 15% Annual Ryegrass 15% Hard Fescue 10% White Dutch Clover 5% Canada Bluegrass 2% Redtop 1%	General seeding inland where mean annual precipitation is >50 cm. Fertilizer: 22-11-11
	Interior Dryland Mix Crested Wheatgrass 40% Tall Wheatgrass 25% Slender Wheatgrass 20% Hard Fescue 15%	General seeding inland where mean annual precipitation is < 30 cm. Fertilizer: 22-11-11
	Alkaline Tolerant Blend Crested Wheatgrass 35% Sherman Big Bluegrass 20% Hard Fescue 20% Canada Bluegrass 15%	General seeding in alkaline soils. Fertilizer: 22-11-11
Kootenays	**Interior Forestland Mix** Intermediate Wheatgrass 32% Alfalfa ("Rambler") 20% Perennial Ryegrass 15%	General seeding inland where mean annual precipitation is >50 cm.

CLIMATIC AREA	STANDARD MIXES (by weight)		APPLICATION
	Annual Ryegrass	15%	Fertilizer: 22-11-11
	Hard Fescue	10%	
	White Dutch Clover	5%	
	Canada Bluegrass	2%	
	Redtop	1%	
	Kootenay Dryland		General seeding inland where mean annual precipitation is < 50 cm. Fertilizer: 22-11-11
	Tall Wheatgrass	45%	
	Crested Wheatgrass	20%	
	Alfalfa ("Rambler")	15%	
	Hard Fescue	7%	
	Sheep Fescue	5%	
	Alsike Clover	5%	
	Canada Bluegrass	2%	
	Redtop	1%	
Northern **(Prince George Area)**	**North East General Mix**		General seeding inland where mean annual precipitation is > 50 cm. Fertilizer: 26-16-8
	Smooth Bromegrass	40%	
	Creeping Red Fescue	20%	
	Timothy	15%	
	Alfalfa	15%	
	Alsike Clover	10%	
	North East Dryland Mix		General seeding inland where mean annual precipitation is < 50 cm. **Fertilizer: 26-16-8**
	Crested Wheatgrass	35%	
	Intermediate Wheatgrass	25%	
	Alfalfa	15%	
	Smooth Bromegrass	10%	
	Creeping Red Fescue	10%	
	Alsike Clover	5%	
Northern **(Terrace Area)**	**North West General Mix**		General seeding inland where mean annual precipitation is > 50 cm. For use in CWH and ICH biogeoclimatic zone. Fertilizer: 22-11-11
	Smooth Bromegrass	70.6%	
	Alfalfa	18.0%	
	Creeping Red Fescue	3.9%	
	Alsike Clover	3.4%	
	Timothy	2.9%	
	Kentucky Bluegrass	1.1%	

CLIMATIC AREA	STANDARD MIXES (by weight)	APPLICATION
	Northern Coastal Mix: Alfalfa 46.6% Intermediate Wheatgrass 14.0% Smooth Bromegrass 10.3% Kentucky Bluegrass 9.0% Sheep Fescue 8.2% Birdsfoot Trefoil 6.2% Timothy 5.7%	General seeding coastal locations where mean annual precipitation is > 90 cm. For use in CWH biogeoclimatic zone (QCI, Prince Rupert to Pacific) Fertilizer: 22-11-11
	Northern Mix: Hairy Vetch 57.3% Crested Wheatgrass 16.4% Alfalfa 13.1% Creeping Red Fescue 5.3% Orchardgrass 3.8% Birdsfoot Trefoil 3.1% Kentucky Bluegrass 1.1%	General seeding coastal locations where mean annual precipitation is > 90 cm. For use in ICH and BWBS biogeoclimatic zones. (ICH – Pacific to Moricetown, Kitwanga to Thomas Creek 220 km N) (BWBS – Thomas Creek to Yukon border) Fertilizer: 22-11-11
	Ditch Vegetation Seed Mixture Crested Wheatgrass 38.7% Alfalfa 30.9% Creeping Meadow Foxtail 15.8% Birdsfoot Trefoil 6.9% Reed Canarygrass 4.8% White Clover 1.6% Kentucky Bluegrass 1.2%	For use in revegetating roadside ditches following ditch maintenance operations. Fertilizer: 22-11-11
Vancouver Island	**Vancouver Island / Coast Mix** Perennial Ryegrass 26% Creeping Red Fescue 24% Alsike Clover 14% Hard Fescue 13% White Clover 9% Timothy 8% Canada Bluegrass 4% Redtop 2%	General seeding coastal locations where mean annual precipitation is > 90 cm. Fertilizer: 18-18-18

MEAN ANNUAL PRECIPITATION



SECTION 766

IRRIGATION

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

766.01 Scope - The work consists of the supply of all material, labour and equipment to install a complete and operating irrigation system as shown on the Design. The Contractor shall be responsible for obtaining all permits required.

766.02 Site Security - The Contractor shall be responsible for maintaining all security at the project Site at all times, and shall ensure that no damage or breakage occurs to stockpiled materials or to the partially installed irrigation system. The Contractor shall make good all damage resulting from acts of vandalism throughout the period of installation and subsequent maintenance.

766.03 Electrical Trades - The Contractor shall contact the Electrical Trades Supervisor prior to performing work on any existing Ministry owned equipment. All AC electrical connections shall be done by a qualified electrician.

766.04 Drawings and Instructions - The Contractor shall install the irrigation system in accordance with the Design and the Contract.

The Contractor shall maintain a daily record of construction activities. Upon completion of the Contract, the Contractor shall incorporate all accumulated information relevant to the Contract into the required as-built Drawings. The as-built Drawings shall be reproducible, and shall be compiled by a competent professional draftsperson.

The Contractor shall prepare drawings showing the final location and make of all heads, emitter locations, pipe layout, electrical information, wiring, controller station numbers, operating instructions, spare parts lists, and other pertinent information.

Completion will not be certified until adjustments and Drawings are approved.

766.05 References - Codes & Standards - In the absence of other instructions, the provisions of all the following codes and standards shall apply: The National Building Code of Canada; Current CSA Specifications for copper, steel and plastic pipe; AWWA Specification, current editions.

MATERIALS

766.11 General - Shipping, handling and installation of materials shall be to manufacturer's recommended instructions, and best work practice. Particular care shall be taken to avoid scratches and nicks on the plastic pipe. Pipe must be properly stacked and stored in a clean place on the Site, keeping dirt out of the pipe at all times.

766.12 Pipes and Fittings - Galvanized pipe, Schedule 40, with galvanized fittings, shall be used inside culverts. The pipe shall be connected to the plastic main or lateral 450 mm clear of the culvert.

Plastic pipe shall be used for the submain and laterals of the irrigation system. Plastic pipe shall be semi-rigid extruded from PVC (Polyvinyl Chloride) resin, Type 1, grade 2, normal impact.

The minimum classes to be used are listed in Table 766-A.

Fittings shall be PVC plastic, Schedule 40 or 80, designed for solvent welding to PVC pipe.

All fittings must have 1/2 to 2/3 interface fit to ensure a fully seated joint. Individual fittings shall be selected to ensure a proper fit or they will be rejected.

All pipe and fittings shall be continuously and legibly marked with at least the following information:

- Manufacturer's name or trademark;
- Pressure rating;
- Type of material.

Pipe that is not marked will be rejected and shall be removed from the Site by the Contractor.

766.13 Cement - Pipe cement for solvent welding shall be of the type and make recommended by the pipe manufacturer, supplied to the Site in sealed containers clearly marked with the name of the manufacturer and the lot number. The Contractor shall comply with the manufacturer's instructions and safety procedures.

766.14 Sprinkler Heads - Risers for turf heads, lawn heads, impact heads, shrub heads and quick-coupling (QC) valves shall be swing joint type, fabricated from Schedule 80 PVC or Schedule 40 galvanised pipe and fittings, as detailed in the Specifications. Teflon tape shall be used on all threaded connections.

Sprinkler heads shall be as detailed on the Design.

TABLE 766-A MINIMUM CLASSES OF PLASTIC PIPE

Up to 25 mm diameter	Class 200 for excavated trenches
30 mm diameter and larger	Class 160 for excavated trenches
All sizes	Class 200 for pipe flow

766.15 Valves - Automatic valves shall be as detailed on the Design and the same make shall be used throughout. Automatic valves shall have flow control stems.

Valves shall be installed in Carson Industries No. 1419 valve boxes or other approved alternative complete with extensions and covers, as required. The top of all valve box covers shall be flush with the finished grade.

766.16 Controllers - Automatic controllers shall be supplied to operate the electrically controlled automatic valves. Controllers shall be 24 volt A.C. outlet, Class 2 rating, compatible with the valves used. Controllers and automatic valves shall be by the same manufacturer, unless otherwise noted. Controllers and transformers must bear CSA or Provincial stamps of approval. Controllers shall be as detailed on the Design.

Controllers shall be installed in Ministry standard controller box 30-A-120/240V, weatherproof, stainless steel service panel SN1765A as shown on SS Drawing SP635-2.4.8, or pre-approved equal complete with a Masterlock No. 15 padlock and two sets of keys for the lock.

766.17 Wiring - Wire between controllers and automatic valves shall be of a type approved for direct burial. Where control wires are exposed or pass through culverts, they shall be installed in rigid electrical conduit.

Wire shall be minimum 14 gauge single strand T.W.U.

766.18 Selected Native Fill - Native fill selected for backfilling shall be free of stones, gravel, wood or any other debris.

CONSTRUCTION

766.31 General - Damaged Material - Damaged material shall not be used. The Contractor shall take care to prevent dirt from entering the pipe.

Plastic pipe shall not be repaired by patching. Where pipe has been damaged, the damaged section shall be removed and a new section shall be installed complete with new fittings.

766.32 Line Location - The Contractor shall ensure that all irrigation pre-ducts for passage of irrigation lines under roadways, medians, traffic islands and other surface impediments have been installed and are clearly marked at all entry points.

No irrigation line shall be installed parallel to and directly over another irrigation line or line of another trade. Lines laid in the same trench shall be a minimum of 50 mm apart. No pipe shall be installed closer than 300 mm to any parallel electric conduit as shown on SS Drawing SP635-1.5.2.

766.33 Compaction - Before laying the pipe, the Contractor shall be satisfied as to the extent of compaction in the lawn and planting areas.

766.34 Excavation - Excavated soil shall be carefully placed adjacent to the trench for convenient backfilling. Topsoil and subsoil shall be piled separately to avoid contamination of the topsoil.

Stones or other objects larger than 75 mm at their widest point shall be removed from the trenches. Holes below grade lines, caused by the removal of stones, must be filled in and compacted uniformly with the adjacent trench.

766.35 Laying the Pipe - Pipe shall be laid by trench excavation or by an approved vibrating pipe plough. Plastic pipe shall be laid on sand or selected native fill to a compacted depth of 50 mm. A further 75 mm of sand or selected native fill shall be placed over plastic pipes prior to backfilling. Pipes shall be run in straight lines between fittings. Pipe must not be supported at intermediate points on stones, bricks or other hard material.

All mains and laterals shall have a minimum cover of 400 mm of soil as measured from the top of the pipe to the finished grade.

Lawn and planting areas shall be disturbed as little as possible.

The Contractor shall manicure the finished grade over all mains and laterals upon completion of the pipe installation. All debris, rocks over 50 mm diameter, etc, that have been brought to the grade surface shall be removed to the Contractor's own tip. The area over all trenches shall be fine graded and shall conform to DBSS 751.34.

766.36 Connections - The Contractor shall make connection to the existing water supply where shown on the Design. The Contractor shall ensure that the recommended operating pressure of the irrigation system is not exceeded by the water pressure at the source, by installing pressure regulators as required.

766.37 Inspection and Testing - After the pipe is in place in the bottom of the trench with risers in place, the risers shall be capped where the sprinklers will be attached and all pipe fittings exposed. The maximum pressure shall be applied to the system and maintained for a minimum of one hour.

All fittings shall be visually inspected and any that leak

shall be cut out and replaced. Leaks shall not be repaired by patching. The test pressure shall be maintained for one hour after replacing any defective sections. The section shall be re-inspected as before.

The system shall be flushed out to remove dirt and then the sprinklers shall be attached using Teflon tape or pre-approved non-setting pipe thread compound.

766.38 Backfill - After approval by the Quality Manager, the trenches shall be backfilled, maintaining pressure in the line. If there is any indication of a leak, the defective section shall be located and replaced.

The trenches shall be carefully backfilled with the subsoil, followed by the topsoil. Both shall be compacted to the same density as the soil in the trench walls to minimize differential settlement. Backfill around turf heads with 0.03 m^3 of sand.

766.39 Controller Installation - The location of the controllers shall be determined on Site in the areas indicated on the Design. The Contractor shall have a qualified electrician connect the controllers to the electrical supply.

766.40 Adjustments - The sprinkler system shall be adjusted section by section to give satisfactory coverage to all areas. Pressure at the heads and/or Q.C. valves shall be as noted on the Design. Turf heads, lawn heads and Q.C. valves shall be set flush with the final turf grade by adjusting the swing joint riser, as required. During the landscape warranty period, the Contractor will return twice and adjust the heads, as required, to be flush with the final turf grade.

These callbacks shall be done within five days of notification by the Ministry Representative and shall be considered part of the Contract requirements.

766.41 Surplus Material - Surplus material shall be removed from the Site.

766.42 Conditions for Acceptance - Completion will not be certified until adjustments are completed, as-built Drawings prepared, and a copy of the inspection certificate issued by the Ministry of Competition, Science and Enterprise indicating compliance with the Electrical Code is provided.

766.43 General - The Contractor shall monitor the operation of the system and carry out all minor repairs and required adjustments to the spray coverage of irrigation heads and operating times.

The irrigation system shall be properly winterized at the appropriate time of the season.

SECTION 769

PROTECTION AND RETENTION OF VEGETATION

DESCRIPTION

769.01 Scope - This Section refers to the protective measures required to safeguard vegetation from construction operations, equipment and vehicles, where vegetation is not designated for removal under the Contract, and covers the installation of barriers.

769.02 Related Work – DBSS 165, Protection of the Environment; DBSS 200, Clearing and Grubbing; DBSS 201, Roadway and Drainage Excavation; DBSS 751, Topsoil and Landscape Grading.

769.03 Definitions

Specimen Trees - means trees so designated in the Contract Documents.

Native Vegetation - means areas of existing and/or indigenous shrubs, trees and groundcover.

Dripline - means the location on the ground surface directly beneath a theoretical line described by the tips of the outermost branches of trees.

Barrier - means fence consisting of approved material, supported by steel posts and being a minimum of 2.0 m high, without breaks or unsupported sections.

MATERIALS

769.11 Water - Water shall be free of impurities that would inhibit germination and growth or may be harmful to the environment.

The Contractor shall supply the water.

769.12 Fertilizer - Fertilizer shall be supplied to the specifications of the Contract and to DBSS 751.18.

CONSTRUCTION

769.31 Operational Constraints - The Contractor's operations shall not damage vegetation designated for retention.

Existing vegetation shown on the Design or designated in the Contract to be retained shall be marked by the Contractor.

Areas designated in the Contract or by the Design as "Vegetation to remain" areas are not to be disturbed, cleared or logged.

The Contractor's operations shall not cause flooding, sediment deposits or deposition of debris in "Vegetation to Remain" areas.

Where construction procedures substantially alter natural drainage patterns, interim drainage or irrigation shall be provided as necessary to compensate for construction interference.

Construction procedures, stockpiling of materials or debris burning or disposal shall not be undertaken adjacent to designated trees and/or native vegetation retained.

Unless the Contract requires work within the dripline of trees designated to remain, equipment shall not be operated within that dripline. When the Contract requires work within the dripline of trees designated to remain, operation of equipment within that dripline area shall be kept to the minimum necessary to perform the work required. Tree roots shall be protected from compaction by temporary placement of hogfuel or other lightweight insulative material, as required by the Designer or Quality Manager.

Equipment or vehicles shall not be parked, repaired or refuelled, construction materials shall not be stored and earth materials shall not be stockpiled within the dripline area of any tree designated to remain.

769.32 Clearing and Grubbing - No clearing and grubbing shall be conducted for a radius of 3 m from the trunks which lie on the edge of clearing and grubbing zones.

Trees to be removed shall be felled toward the centre of an area, away from selectively cleared or retained vegetation.

769.33 Barriers for Existing Vegetation Protection - Barriers for vegetation protection shall be erected prior to commencement of construction operations, at locations specified in the Contract or Design, to provide a continuous barricade between vegetation and the area of work. The barriers shall be maintained erect and in good repair throughout the duration of construction operations, and shall be removed upon completion of the work, and disposed of outside the project by the Contractor.

The barrier shall be placed at the dripline of trees or forest edges unless this is inadequate space to provide a 1.5 m buffer zone between the barrier and the limit of grading. The barrier shall be placed within the dripline if necessary to provide a buffer zone of up to 1.5 m. Under no circumstance shall it be placed less than 0.75 m from the circumference of the trunk. When the trunks of trees are

less than 4.5 m apart, the trees shall be considered a group, and the barrier shall be placed to form a continuous barricade.

A barrier is not required where an existing fence will serve the same purpose. At such locations, the barrier shall terminate at the existing fence so that a continuous barricade is provided between the trees and the area of work.

769.34 Pruning and Repair of Specimen Trees - Specimen trees and trees safeguarded by barriers shall be repaired as follows..

One third of the tree branches shall be selectively removed to reduce transpiration and compensate for dieback of roots in fill conditions and damage to the root system in cut conditions.

Within five calendar days of damage, branches 25 mm or greater in diameter that are broken as a result of the Contractor's operations shall be cut back cleanly at the break, or to within 10 mm of their base, if a substantial portion of the branch is damaged.

Roots 25 mm or larger in diameter that are exposed by the Contractor's operations, shall be cut back cleanly to the soil surface within five calendar days of exposure.

Bark that is damaged by the Contractor's operations shall be neatly trimmed back to uninjured bark, without causing further injury, within five calendar days of damage.

MAINTENANCE

769.71 Watering - The retained specimen trees shall be watered three times during the summer or as stated in the Contract or by the Design. The area immediately below the tree crown shall be soaked sufficiently to reach the feeder roots.

769.72 Fertilizing - Where specified, fertilizer shall be applied at a rate of 50 gr/mm of calliper to existing specimen trees to be retained. The calliper measurement shall be taken 0.3 m above the grade. The fertilizer shall be applied once early in the growing season unless specified otherwise.

SECTION 903

TIMBER - MATERIALS

903.01 General - All timber shall be graded in accordance with the current Standard Grading Rules of the National Lumber Grades Authority. Timber must be grade stamped with the exception of unfinished or rough timber, which may prove too difficult to stamp, in which case a grading certificate may be requested. Timber materials may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

903.02 Species and Grade - Timber species and Grade shall be as specified on the Design and in accordance with the Contract.

903.03 Inspection - All timber may be inspected before shipment to the construction site, or at the construction site, or both. The Contractor shall provide the necessary facilities to enable the Quality Manager and Ministry Representative to expeditiously examine as many pieces as are deemed necessary. All material rejected shall be replaced. Inspection of timber before shipment shall not be a bar to its subsequent rejection at the construction site if found to fail any requirements of this specification.

No material shall be shipped prior to inspection or until a release for shipment has been issued by the Quality Manager.

SECTION 904

TIMBER (TREATED AND UNTREATED) - FABRICATION AND HANDLING

904.01 General - All timber shall conform to the requirements of DBSS 903, Timber - Materials. Timber to be treated shall be treated in accordance with DBSS 908, Preservative Treatment - Wood Products. Except as modified herein, preservative treated wood products shall be handled in accordance with CSA Standard 080-M4.

Timber materials may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

904.02 Framing - All the cutting, boring, framing, match marking, etc. required on all timber shall be done by competent bridge framers in a thorough manner, in accordance with good work practice. It shall be done such that surfaces in contact shall bear evenly and fully; no shims or open joints are permitted. All measurements shall be accurate. Gains and daps shall have plane, smooth surfaces.

904.03 Bridge Iron - The Contractor will supply all iron that is necessary for use in the framing operations. Materials shall conform to the requirements of DBSS 911.

904.04 Handling of Untreated Timber - All materials shall be handled with reasonable care. Timber shall not be bashed, bruised, gouged, cracked, split or otherwise damaged. Minor damage, in the opinion of the Quality Manager and Ministry Representative, shall be repaired; severely damaged materials will be rejected and replaced.

904.05 Preparation of Timber Before Treatment - All cutting, such as boring, chamfering, framing, gaining, surfacing, trimming etc., shall be done prior to treatment.

In the event that cutting becomes absolutely necessary after treatment, the cut surfaces shall be saturated with preservative according to CSA Standard 080-M4 Paragraph 1.5.

904.06 Condition After Treatment - After treatment, timber with checks exceeding the limiting sizes specified for the appropriate grade shall be rejected (see DBSS 903, Timber-Materials).

904.07 Handling of Treated Timber - Pointed tools or equipment such as dogs, hooks, peavies, etc. shall not be used on the side surfaces of treated timber. The use of pointed tools shall be confined to the end grain and shall be such as to avoid damage to the pressure-treated surface.

Treated timber shall not be dragged along any surface.

SECTION 905

TIMBER - GLUED LAMINATED

905.01 General - The requirements of the current CSA Standard 0122 - Specification for Glued Laminated Softwood Structural Timber shall apply to all glued laminated members. Wood species shall be as specified on the Design, except that if the members are to be pressure treated in accordance with CSA 080-28 (See DBSS 908) they may only be of Coast Region Douglas Fir or Western Hemlock.

Timber materials may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

905.02 Classification - Unless otherwise specified in the Contract and the Design, the appearance grade shall be Industrial and the service grade shall be Exterior. The stress grade shall be as shown on the Design.

905.03 Quality Control - Laminations shall be grade-marked and the marking shall be visible until the glue has been applied.

The shear tests described in Appendices "A" and "B" of CSA Standard 0122 shall be carried out by the Quality Manager. Copies of the shear test results shall be forwarded to the Ministry Representative if requested.

Vacuum-pressure cycle tests may be carried out by the Quality Manager, and the Contractor shall supply the Quality Manager with a full section cut-off, 75 mm long, from each end of the member for this purpose.

905.04 Incising - The end bearing surfaces of members shall be incised prior to pressure treatment.

905.05 Handling of Treated Glued Laminated Members - Members shall be protected to avoid damage due to handling: dogs, hooks, peavies or other equipment shall not be used on the side surfaces of treated timber. All handling of treated timber with pointed tools shall be confined to end grain and shall be such as to avoid damage to the original pressure-treated surface.

Subject to the approval of the Quality Manager and Ministry Representative, the Contractor shall make good superficial damage of treated timber by the methods outlined in the current edition of CSA Standard 080 - Wood Preservation. Timber which, in the opinion of the Quality Manager or Ministry Representative, cannot be made good by such methods will be rejected.

905.06 Storage of Glued Laminated Members - Members lying in storage shall be supported evenly on a flat surface. When stored for a prolonged time, they shall be gapped to permit air circulation.

905.07 Inspection - All timber and all phases of the work including pressure treatment, if applicable, may be inspected by the Quality Manager or the Ministry Representative. The Quality Manager shall be given 48 hours notice of commencement of gluing and pressure treating, if applicable. The Contractor shall provide the necessary facilities to enable the Quality Manager or the Ministry Representative to expeditiously examine as many pieces as are deemed necessary by the Quality Manager or the Ministry Representative. All material rejected shall be replaced. Inspection of glued laminated timber before shipment shall not be a bar to its subsequent rejection at the Site if found to fail any requirements of the Contract.

No material shall be shipped prior to inspection or before a release for shipment has been issued by the Quality Manager.

SECTION 906

ROUND TIMBER PILES

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

906.01 Scope - This Section covers the quality and manufacture of untreated and pressure-treated round timber piles.

Round timber piles may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

Where convenient, the requirements contained in this Section use the wording of the current CSA Standard CAN3-056-M "Round Wood Piles", in particular for term definition (see Appendix A) and the majority of measurement and material requirements.

Preservative treatment of piles shall conform to the requirements of DBSS 908, Preservative Treatment - Wood Products.

906.02 Species

906.02.01 Untreated Piles - This specification does not restrict the species used for untreated piles except as may be specified in the Design or in the Contract.

906.02.02 Pressure Treated Piles - Species of piles to be pressure treated shall be as specified in the Design and in the Contract and restricted to those for which pressure treating specifications are included in CSA Standard 080, Wood Preservation. Pressure treatment of piles shall be according to DBSS 908.

906.03 Size - The size of a pile shall be designated by length and either minimum and maximum butt diameter or minimum and maximum tip diameter, or both. Sizes of piles normally available are shown in Table 906-A.

906.04 Diameter

a) All measurements of diameter shall be made inside the bark.

b) The diameter of treated piles shall be measured after treatment.

c) Butt diameters shall be measured at the extreme butt.

d) Tip diameters shall be measured at the extreme tip.

e) The diameter of a pile shall be determined either by measuring the circumference in millimetres and dividing the result by 3.14 or by taking the average of the maximum and minimum diameter.

f) A variation of -15 mm in the diameter at the tip or at the butt but not in both, shall be allowed in not more than 25% of the piles when the specification of four or more piles is to be determined.

g) If allowances are specified on the order, they shall

TABLE 906-A SIZES OF TIMBER PILES

Size Designation	36	33	30	27	24
Minimum Diameter at Extreme Butt or Large End (mm)	360	330	300	270	240
Length in Metres	**Minimum Diameter at Small End Tip (mm)**				
Up to 6	250	250	230	200	180
6 to 11	250	230	200	180	150
12 to 14	230	200	180	150	-
15 to 18	200	180	180	-	-
19 to 21	200	180	150	-	-
22 to 27	180	150	-	-	-
28 to 32	150	130	-	-	-

Note: Diameters are minimum except for tolerance permitted in DBSS 906.04 (f). Maximum diameter at butt shall not exceed 500 mm for any pile size.

supersede DBSS 906.04 (f).

906.05 Length

a) Pile lengths shall be measured in metres.

b) A variation of ±150 mm from designated lengths will be permitted, unless allowable over and under lengths are stated on the order.

906.06 General Material Requirements - Piles shall be cut from sound live trees. Sides and end surfaces of the piles shall be free of ice, snow and mud, and exposed for visual inspection.

Untreated piles shall be sound and close-grained.

Treated piles shall:

- have not less than 25 mm sapwood, and shall be cleanly peeled;
- not be cored until 24 hours after treatment;
- be cored in the middle third of the length of the pile.

906.07 Prohibited Defects

- Through checks
- Bird holes
- Cross-breaks of grain (cracks)
- Decay
- Nails, spikes and other metal or foreign substance.
- Holes in treated piles (except holes for test purposes, which shall be properly plugged). Holes for brailing purposes shall be drilled prior to treatment.
- Shakes in the tip.
- Splits in the tip
- Insect damage
- Any abnormal change in cross-section (including ground swell)
- Knot clusters
- Unsound scars (cat faces)
- Short crooks (see Drawing SP906-01)
- Reverse sweep (see Drawing SP906-02)
- Burst - unsound piling which have 15 mm or more of a concentration of oil in the deadwood.

906.08 Permitted Defects

- Firm red heart
- Hard stain
- Compression wood

906.09 Limited Defects

a) Checks caused by treating and normal seasoning shall not exceed the following:

i) Checks in the tips of the pile from the pith to the circumference shall not be wider than 10 mm at the circumference.

ii) If checks are wider than 10 mm but not deeper than 1/3 of the diameter of the pile, not more than three of these checks shall occur in the circumference of the pile at that point. If checks are deeper than 1/3 of the diameter of the pile but not to the heart, and not wider than 10 mm, then only two such checks shall occur in the circumference of the pile at that point. The sum of all the widths of checks in a pile shall not exceed 1/4 of the diameter where they occur.

b) Sound Knots will be permitted as follows:

i) For piles 15 m and less in length, knots up to 100 mm in diameter will be permitted provided that the sum of all knot diameters does not exceed 1/3 of the diameter of the pile at the section where they occur;

ii) For piles more than 15 m in length, knots between:

- mid-length and butt shall conform to the requirements for piles of 15 m or less;
- mid-length and the tip, up to 120 mm in diameter, will be permitted provided that the sum of all knots does not exceed 1/2 of the diameter of the pile at the cross-section where they occur.

c) Shakes in the butt end of a pile are permitted provided that they are not closer than 50 mm to the surface of the pile and the length of any single shake, or the total length of any number of shakes measured along the line of the shake does not exceed 1/3 of the measured diameter of the butt.

d) Splits in the butt end of a pile are permitted provided that they are not longer than the diameter of the butt.

e) Sweep is permitted provided that it cannot be classified as short crook or reverse sweep and provided that:

i) For piles less than 20 m in length, a straight line joining the mid-point of the butt and the mid-point of the tip does not at any intermediate point pass outside the surface of the pile;

ii) for piles 20 m and up to 24 m in length, a similar straight line does not lie more than 30 mm outside the surface of the pile; or

iii) for piles over 24 m in length, a similar line does not lie more than 60 mm outside the surface of the pile.

Note: See SS Drawing SP906-01

f) Holes for untreated piles less than 10 mm in average

diameter will be permitted provided that:

i) they are scattered over the surface of the pile; and

ii) the sum of the average diameter of all holes in any square 300 mm x 300 mm of pile surface does not exceed 40 mm and the depth of any one hole does not exceed 40 mm.

g) Spiral Grain shall not exceed 1/2 turn in any 6 m length of pile.

906.10 Manufacturing Requirements - All piles shall be cut above the ground swell and have a uniform taper throughout their whole length.

Completely overgrown knots rising more than 30 mm above the pile surface, branch stubs, and partially overgrown knots shall be trimmed close.

Peeling of piles by hand or machine shall be done carefully so as not to gouge or otherwise damage the surface of the pile, and the reduction in thickness of sapwood shall be the minimum possible.

Sawing of butts and tips shall be cut perpendicular to the axis of the pile. Bevelling at the ends of the piles shall not be permitted.

906.11 Storage - Piles shall be stacked in treated or other non-decaying skids of such dimensions, and so arranged as to support the piles without producing noticeable distortion of any of them. A cover should be put over them to protect against the elements.

906.12 Handling - Handling damage will be unacceptable if it reduces the depth of sapwood so as to render the pile untreatable.

a) All piles shall be handled with reasonable care. Piles shall not be bruised or otherwise damaged. Minor damage shall be repaired and severely damaged piles shall be replaced as per DBSS 906.12 (d).

b) During lifting, long piles shall be supported at a sufficient number of properly located points to prevent damage due to excessive bending.

c) Dogs, hooks, peavies or other equipment shall not be used on the round surface of treated piling, except in the last one metre of the butt end.

d) The Contractor shall make good any superficial damage of treated piles by methods outlined in the current edition of CSA 080 "Specification of Wood Preservation". Piles, which, in the opinion of the Quality Manager and Ministry Representative, cannot be made good by such methods, will be rejected.

906.13 Inspection - All piles may be inspected before shipment to the Site, at the Site or both. The Contractor shall provide the necessary facilities to enable the Quality Manager, Ministry Representative or authorized Inspector to expeditiously examine all parts of each pile. All piles rejected shall be replaced,. Inspection of the piles before shipment shall not be a bar to their subsequent rejection at the Site if found to fail any requirements of this Section.

The Quality Manager and Ministry Representative shall be informed by the Contractor in sufficient time before treatment, and, in all cases, before shipment to the Site, so that necessary inspection may be carried out.

Treated piles being supplied from previously treated stock shall have been clearly stamped with the applicable Charge Number or other positive identification, and the appropriate treatment records shall be made available to the Quality Manager and, if requested, the Ministry Representative for inspection. All piles being supplied from previously treated stock shall be subject to re-inspection.

Piles to be supplied from previously treated stock which have been stored in water shall, if ordered by the Quality Manager or Ministry Representative, be removed from the water if such action is required to provide adequate inspection. At the Contractor's option, such inspection may be carried out at the Site. All piles rejected at the Site shall be removed and replaced at the Contractor's expense.

906 APPENDIX A - Definitions

Check: separation of the wood along the grain, the greater part of which occurs across the rings of annual growth.

Through check: a check, which extends from surface to surface of the pile, usually through the pith centre.

Compression wood: wood which has grown abnormally as often occurs on the lower side of branches and inclined trunks of coniferous trees.

Compression wood:

- is denser and harder than normal wood but may be lower in strength for its mass.
- is characterized by relatively wide annual rings, usually eccentric.
- has a relatively high proportion of summerwood (frequently more than 50% of the width of the annual rings in which it occurs).
- exhibits little contrast in colour between springwood and summerwood, and
- shrinks excessively lengthwise as compared with normal wood.

Crack: separation of the wood cells across the grain (this may be due to internal strains resulting from unequal longitudinal shrinkage or to external forces).

Decay (rot, dote): the disintegration of the wood substance, due to the action of wood-destroying fungi (rot and dote mean the same as decay).

Grain: the direction, size, arrangement, and appearance of fibres in a pile.

Spiral grain: a type of growth in which the fibres take a spiral course about the bole of a tree instead of the normal vertical course. The spiral may extend right-handed or left-handed around the tree trunk. The amount of spiral grain in a pile is measured as the distance in metres, along the axis of the pile in which one complete twist of the spiral occurs, and is expressed as, for example, 1 turn in 12 m.

Hole: an opening, perforation or cavity in a pile.

Bird holes: holes and damage caused by woodpeckers and other species of birds.

Plugged holes: holes that have been filled by forcing in another piece of wood.

Insect damage: result of boring in the pile by insects or insect larvae. Scoring or channelling of the surface is not classified as insect damage.

Knot: that portion of a branch or limb that has become incorporated into the body of a tree.

Knot cluster: two or more knots grouped together as a unit with the fibres of wood deflected around the entire unit.

Knot diameter: the diameter of a knot as it appears on the surface of a pile measured in a direction at right angles to the lengthwise axis of the pile.

Unpeeled: no attempt is made to remove the bark from the pile, but does not mean that all of the bark is on the pile as it may be scuffed, knocked, or worn off after the pile is cut.

Rough peeled: all of the outer bark is removed from the pile.

Clean peeled: all of the rough bark is removed from the pile, and all of the inner bark from at least 80% of the surface of the pile, no piece of inner bark more than 200 mm long remains on the surface of the pile, and there is at least 30 mm of clean wood surface between any two strips of inner bark.

Red heart: a condition caused by fungus (Fomes pini, Trametes pini) which occurs in the living trees of some conifers. It is characterized in the early stages of infection by a reddish or brownish colour in the heartwood. Later the wood in the living tree disintegrates (decays) in small, usually distinct areas that develop into white-lined pockets.

Firm red heart: the early stages of infection, characterized by a reddish or brownish colour in the heartwood, which does not affect the strength of the pile.

Scar (cat face): a depression in the surface of the pile resulting from a wound where healing has not re-established the normal cross-section of the pile.

Shake: a separation along the grain, the greater part of which occurs between the rings of annual growth.

Short crook: a localized deviation from straightness which, within any section 2 m or less in length, is more than 70 mm (see Drawing SP906-01).

Sound: free from decay and insect holes.

Split: a lengthwise separation of the wood due to the tearing apart of the wood cells which usually extends from surface to surface of the pile.

Stain: a discolouration on or in the wood other than its natural colour, caused by the action of certain moulds and fungi.

Hard stain: a stain, which is not accompanied by softening or other disintegration of the wood.

Sweep: the deviation of the pile from straightness (see Drawing SP906-01).

Reverse sweep: a deviation from straightness, which changes direction or reverses direction in the length of the pile (see Drawing SP906-02).

SECTION 908

PRESERVATIVE TREATMENT - WOOD PRODUCTS

908.01 Materials - Timber, piles and glued laminated members shall conform to the requirements of:

DBSS 903, Timber - Materials; DBSS 904, Timber (Treated and Untreated) - Fabrication and Handling; DBSS 905, Timber -Glued Laminated; DBSS 906, Round Timber Piles as applicable.

Preservative shall conform to the requirements of the current CSA Standard 080-M, "Wood Preservation".

Timber materials may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

908.02 Treatment - All aspects of preservative treatment shall conform to the requirement of CSA Standard 080.1-M "Preservative Treatment of All Timber Products by Pressure Processes".

The type of preservative, conditioning, treatment, penetration and retention shall be appropriate for the species, size and end use of the product, and shall be as specified in the Design and in the Contract.

Note: The type of preservative, treatment and retention will normally be selected by reference to Table 908-A as applicable.

Should it be necessary to use species, commodities or end uses not included in the above standards, then the preservative, treatment and retention shall be selected by reference to CSA Standards 080.4-M, 080.6-M, 080.9-M, 080.11-M, 080.15-M, 080.16-M, 080.25-M, as applicable.

Glue laminated beams shall be treated in accordance with CSA Standard 080.28-M "Preservative treatment of Coast Region Douglas Fir and Western Hemlock structural glued - laminated members and laminations before gluing by the pressure processes." (See DBSS 905.)

Fence posts shall be treated in accordance with CSA Standard 080.5-M "Preservative treatment of posts by pressure processes."

Preservation treatment by the thermal process shall be done in accordance with and to the requirements of current CSA Standard 080-M, "Wood Preservation", and shall be limited to the products and species covered in CSA 080.7-M, 080.8-M and 080.10-M.

908.03 Inspection - The Quality Manager and Ministry Representative shall be given a minimum of 48 hours notice prior to commencement of the treating process. All facilities and reasonable assistance shall be afforded by the supplier, free of cost, to the Quality Manager and the Ministry Representative for the proper execution of the work. They shall have free entry at all times while Work is being performed, to all parts of the treating plant which concern the treatment (and all related work) of the materials ordered.

No material shall be shipped prior to inspection or a release for shipment has been issued by the Quality Manager.

TABLE 908-A TYPE OF PRESERVATIVE, TREATMENT AND RETENTION

CSA Standard 080.14 - M	Pressure preserved wood for highway construction. Table I.	
CSA Standard 080.3 - M	Preservative treatment of piles by Pressure Processes.	
	Table I	Land and Fresh Water Piles
	Table II	Marine Piles
	Table III	Foundation Piles
CSA Standard 080.2 - M	Preservative treatment of lumber, timber, bridge ties, and mine ties by pressure processes. Table I.	
CSA Standard 080.18 - M	Pressure treated piles and timbers in marine construction.	
	Table I	Piles and Timbers
	Table II	Timber Substructure and Superstructure

SECTION 909

TREATED WOOD FENCE POSTS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

909.01 General - This Section covers the quality and manufacture of wood fence and gateposts, braces and droppers.

Wood posts and the like for fences and gates shall be supplied pressure treated in the sizes, species and grading all as required by the Purchase Order, Work Order, Contract, Drawing or Specifications in conformity with this Section and as generally shown on the applicable SS SP741 Drawings.

Note: CSA Standards may be obtained from:

Canadian Standards Association, Standards Sales,
178 Rexdale Road, Rexdale, ON M9W 1P3

909.02 Species - Round fence posts (including line, straining, corner, intersection, gate and end posts) and any required round wood braces and droppers shall be Lodge Pole Pine (Western Jack Pine). Use of other species such as cedar, either split or whole, will not be acceptable.

909.03 Size and Measurement

a) Round fence posts and braces shall be classified as to size on the basis of the smaller diameter and length. The diameter specified shall be minimum with a tolerance of +20 mm and the length shall not vary by more than 25 mm from that specified.

b) The maximum permitted taper shall be such that the larger diameter shall not exceed that of the specified diameter by more than 25 mm over a 2 m length of post or brace. The maximum permitted taper shall be proportional for posts and braces of lengths other than 2 m.

c) All dimensions shall apply inside the bark and to the fully seasoned and treated fence posts or braces.

d) In cases where the post or brace is not truly circular, the diameter shall be obtained by measuring the circumference and dividing by 3.14.

909.04 Prohibited Defects

All round fence posts and braces shall be free of the following defects:

- Decay
- Shakes in the top or butt
- Cracks, splits, through checks
- Spike knots and knot clusters
- Unsound scars
- Reverse sweep

909.05 Limited Defects

All round fence posts and braces are permitted limited defects as follows:

- Seasoning checks less than 10 mm in width if less than 500 mm in length, or less than 5 mm in width if more than 500 mm in length
- Sound scars permitted except within 150 mm from the ends
- Sound, tight, well-spaced knots permitted provided they do not exceed 35 mm in diameter
- Sweep less than 2% of the length of the post
- Short crook less than 2% of the length of the post
- Insect damage consisting of holes 1.6 mm or less in diameter and surface scoring or channelling are permitted. All other forms of insect damage are prohibited.

Note: Defects listed above are as defined in DBSS 906 - Appendix A.

909.06 Manufacturing Requirements - Round Fence Posts and Braces - Ends of posts and braces shall be cut square to the specified length and unless otherwise specified on the Purchase Order, Work Order, or Drawings, one end of posts shall be machine pointed before treatment to permit driving of posts. The diameter at the point shall be not less than 18 mm and not more than 35 mm, and the taper shall extend over a length of 150 mm to 200 mm of the post, or opposite sides shall be tapered to a "chisel point" of similar basic dimensions.

Posts and braces shall be clean peeled with minimum removal of sapwood for their full length and all rough bark and inner bark removed.

All knots or projections shall be shaved smooth and flush with the surface of the surrounding wood.

All round fence posts and braces shall be air seasoned in accordance with CSA Standard 080-M1 Paragraph 1.31 to a moisture content of 15% to 22% before pressure treatment. Where it can be demonstrated that the subsequent pressure treatment performance will be comparable, the Designer or Quality Manager may permit conditioning by steaming for a total of not more than six hours at a temperature not in excess of 115°C.

909.07 Sawn Lumber Posts and Braces - Gate posts and braces not stipulated as round, together with any anchor cleats shall be Standard and better S4S Lodgepole Pine/Western Jack Pine or Coast Douglas Fir, to the current N.L.G.A. grading rules and in the required lengths and sizes.

909.08 Fence Droppers Wood droppers to stabilize barbed wire fencing (Type C) between posts may be:

- Round Fence Posts Lodgepole Pine/Western Jack Pine of 50 mm (smaller) diameter, or
- 25 mm x 50 mm Sawn Lumber, Standard and better S4S Lodgepole Pine/Western Jack Pine or Coast Douglas Fir, to the current N.G.L.A. grading rules.

All droppers 1100 mm ± 25 mm long for 4-wire Type C fences and 1200 mm ± 25 mm for Type C and C2 fences shall be selected for freedom from knots and other imperfections injurious to strength.

Approved proprietary grooved wood droppers with necessary wire clips may be specified or approved for high-tensile smooth-wire fencing.

Note: For Type C fencing proprietary prefabricated galvanized sheet metal and clip droppers may be approved as an alternative to the above wood droppers. Galvanized twisted wire fence stays may be specified or permitted only for the extension of normal fence heights to deer height and the like.

909.09 Pressure Treatment - All round fence posts, braces and round droppers shall be pressure treated in accordance with CSA Standard 080.5 "Preservative Treatment of Posts by Pressure Processes", except that the sample zone for assay shall be 0 to 16 mm from the surface and the requirement that all borings for use in the extraction sample shall have at least 25 mm of sapwood shall be waived. If species other than Lodgepole Pine are approved, these exceptions may be altered.

All sawn fence posts, braces, anchor cleats and droppers shall be pressure treated in accordance with CSA Standard 080.2 "Preservative Treatment of Lumber, Timber, Bridge Ties and Mine Ties by Pressure Processes".

The preservatives and retention of preservatives shall conform to the recommendations of CSA Standard 080.14 Table I "Minimum Retention of Preservatives in Pressure Treated Wood for Highway Construction" under the heading "Fence Posts - All Species" as listed in Table 909-A, and meet the requirements of the relevant CSA Standards in the P-Series.

TABLE 909-A PRESERVATIVES AND RETENTION OF PRESERVATIVES

PRESERVATIVE	MINIMUM RETENTION
Creosote	80.0 kg/m^3
Creosote-Petroleum	95.0 kg/m^3
Pentachlorophenol (Hydrocarbon Solvent) (Type A, or Type D - Methylene Chloride)	4.0 kg/m^3
Ammoniacal Copper Arsenate (ACA)	6.4 kg/m^3
Chromated Copper Arsenate (CCA)	6.4 kg/m^3

909.10 Inspection - All processing of the material shall be documented through the Supplier's QM Plan and sampled through the Contractor's QM Plan. Ministry and Contractor inspectors and auditors shall have free entry to the treating plant while the work is being performed.

No material shall be shipped prior to inspection or the written release for shipment by the Contractor

Material inspected before shipment shall not bar its subsequent rejection after delivery if found to fail any requirements of this Specification. Rejected material shall be replaced at the Supplier's expense including shipping charges and removal of rejected materials, if applicable.

Note: Inspection of material already in bundles ready for shipment shall be considered as "incomplete", and the material will be subject to final inspection only when the bundles are opened immediately prior to use.

The Supplier shall advise the Contractor, within timelines determined by the Contractor, before the material is ready for inspection prior to shipment.

SECTION 911

STEEL AND IRON

911.01 Scope – This standard only applies to materials used in the maintenance and rehabilitation of existing log and timber bridges and log or timber components such as log crib retaining walls, timber piers, wood sign structures, etc.

911.02 Materials - These materials shall be in accordance with the current editions of the following Specifications:

- Structural Steel, CSA Specification CAN/CSA-G40.21 Grade 260 or better
- Steel Bolts, ASTM Specification A 307
- Steel Nuts, ASTM Specification A563
- Carbon Steel Castings, ASTM Specification A 27/A 27M
- Iron Castings, ASTM Specification A 48/A 48M
- Pipe Steel, ASTM Specification A 53/A 53M

911.03 Tension and Lateral Rods - Tension and lateral rods shall be of structural steel. Upset ends shall be upset by hand or machine and welds will not be allowed. The dimensions of upset ends shall be made as shown on the plans. When upsetting by machine, the ends shall be upset to a little oversize, after which they should be heated to welding temperature and then swedged to the proper size. The nuts shall be made hexagonal, U.S. Standard sizes and threads. The threads on rods and nuts shall be full, smooth, uniform and of the same pitch throughout. They shall be such that the nuts can be run on by hand the full length of the threads on the rods without showing undue or uneven slackness.

911.04 Machine Bolts - Machine bolts shall conform to ASTM Standard A 307.

911.05 Carriage Bolts - Carriage bolts shall conform to ASTM Standard A 307. They shall be of the common type with button head, square neck and square or hex nut.

911.06 Drift Bolts or Pins - Drift bolts or pins shall be of structural steel. They shall be cut from plain, round bars unless otherwise called for on the Design. The ends of drift pins or bolts shall be tapered or shaped so that the pin may be easily driven into pre-bored holes in the wood being fastened.

911.07 Dowels - Dowels shall be of structural steel. They shall be cut from plain round bars and any ragged ends shall be removed.

911.08 Lag Screws - Lag screws shall conform to ASTM Standard A 307. They shall have square heads and cone points.

911.09 Plate Washers or Wrought Washers - Plate washers or wrought washers shall be of structural steel or wrought iron; they shall be round or square as may be called for on the Design. Round washers shall be according to Canadian Manufacturer's Standards. Square washers shall be made to the dimensions called for on the Design.

911.10 O.G. Washers - O.G. washers shall be of cast iron. The proportions of these washers adopted by some of the manufacturers differ somewhat in detail, but to be acceptable the diameter should not be less than four times the diameter of the bolt for which it is to be used and the thickness should be approximately equal to the diameter of the bolt.

911.11 Special Castings - Special castings shall be true to pattern, free from excessive shrinkage or overrun, be made in accordance with the Design and be free from defects. Castings, which show defects after machining, will be rejected notwithstanding any previous acceptance at the manufacturer's works.

911.12 Steel Splice-Joint Fastenings - Plates and bars, etc., for these joints shall be of structural steel. All parts shall be made in accordance with the Design. Bolt holes shall be drilled from the solid or sub-punched and reamed. In sub-punched and reamed work, the holes shall be punched 3/16" smaller and after assembling, reamed 1/16" larger than the nominal diameter of the rivet.

911.13 Galvanizing – Galvanizing, if required, shall be in accordance with CSA Specification CAN/CSA-G164.

911.14 Plates - The various plates designated on the plans as gib, bearing, bed, joint, lateral, etc., shall be of structural steel.

911.15 Welding - Welding shall conform to the requirements of CSA Specification W 59.

911.16 Pipe Fastenings - Pipe fastenings shall be of extra strong (Schedule 80) pipe. Nuts shall be hexagonal machine lock nuts, U.S. Standard as to outside dimensions and threads. The threads on nuts and pipes shall be full, smooth, uniform and of the same pitch; they shall be such that they can be run on by hand the full length of the thread on pipe without undue or uneven slackness.

911.17 Cleaning and Painting – Painting of steel and iron components shall be in accordance with DBSS 216.11, Painting of Steel Structures. No paint shall be applied to surfaces which are to be encased in concrete.

911.18 Machine Finished Surfaces - Machine finished surfaces and threaded ends of tension and lateral rods shall be coated with a mixture of white lead and tallow, applied hot, as soon as the surfaces and threads are finished and accepted by the Quality Manager.

911.19 Inspection - All materials and quality of work shall be subject to inspection by the Quality Manager. The manufacturer shall allow the Quality Manager and the Ministry Representative free access to the shops at all times when work on the materials is being done and the manufacturer shall provide every reasonable facility to assist the Quality Manager in the inspection of both material and quality of work. The Quality Manager shall have power to reject material or quality of work which do not come up to the requirements of the Design or the Contract. Rejected material or poor quality of work shall be replaced promptly or made good by the manufacturer. Tests will be made only when and as specified by the Designer or the Quality Manager.

911.20 Shipping - The threaded ends of all rods shall be protected from damage by wrapping with burlap or equally effective covering.

SECTION 913

REINFORCEMENT FOR CONCRETE

913.01 Requirements - Concrete reinforcement shall conform to the requirements of the following CSA Standards:

- G30.3 ..... Cold-Drawn Steel Wire for Concrete Reinforcement
- G30.4 ..... Fabricated Deformed Steel Bar Mats for Concrete Reinforcement
- G30.5 ..... Welded Steel Wire Fabric for Concrete Reinforcement
- G30.14 ..... Deformed Steel Wire for Concrete Reinforcement
- G30.15 ..... Welded Deformed Steel Wire Fabric for Concrete Reinforcement
- G30.17 ..... Welded Steel Plain Round Bar or Rod Mats for Concrete Reinforcement
- G30.18-M .... Billet Steel Bars for Concrete Reinforcement

The type and grade required shall be as specified in the Contract and as shown in the Design.

SECTION 917

WIRE ROPE

917.01 Wire Rope - Wire Rope shall conform to the requirements of CSA Standard G4-M "Steel Wire Rope for General Purpose and for Mine Hoisting and Haulage."

SECTION 931

PORTLAND CEMENT

931.01 General- Portland Cement shall conform to the requirements of CAN/CSA-A3001, "Cementitious Material for use in Concrete"

The Type shall be as specified in the Contract and as shown in the Design and as required from Table 931-A.

TABLE 931-A TYPES OF PORTLAND CEMENT

NAME	TYPE	APPLICATION
Normal	GU	For use in general concrete construction when the special properties of the other types are not required.
Moderate*	MS	For use in general concrete construction when moderate resistance to sulphate action is required..
Moderate**	MH	For use in general concrete construction when moderate heat of hydration is required.
High Early Strength	HE	For use when high early strength is required.
Low Heat of Hydration	LH	For use when low heat of hydration is required.
Sulphate Resistant	HS	For use when high sulphate-resistance is required.
* Moderate with respect to sulphate-resistance. ** Moderate with respect to heat of hydration.		